

Natural mood foods: The actions of polyphenols against psychiatric and cognitive disorders

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Objectives Polyphenols, natural compounds found in plant-based foods, possess special properties that can battle oxidative stress and stimulate the activation of molecules that aid in synaptic plasticity, a process that underlies cognitive function. Unlike many traditional treatments, polyphenols affect a broad range of mechanisms in the brain that can assist in the maintenance of cognitive and mental health, as well as the recovery from neurodegenerative diseases. Examining the molecular basis underlying the link between food intake and brain function has presented the exciting possibility of using diet as a viable method to battle cognitive and psychiatric disorders.

Methods We will discuss the molecular systems that link polyphenols, the gut, and the brain, as well as introduce published human and animal studies demonstrating the effects of polyphenol consumption on brain plasticity and cognition.

Results By influencing cellular energy metabolism and modulating the signaling pathways of molecules involved with brain plasticity, dietary factors – formerly recognized for just their effects on bodily systems – have emerged as affectors of the brain.

Conclusion Thus, the consumption of diets enriched with polyphenols may present the potential of dietary manipulation as a non-invasive, natural, and inexpensive therapeutic means to support a healthy brain.